MILITARY SPECIFICATION

VOLTAGE REGULATOR SUBASSEMBLY: 10525755

1. SCOPE

1.1 This specification covers the Voltage Regulator Subassembly: 10525755 which contains the control elements of the plus 150 volt voltage regulator, the plus 15 volt voltage regulator, and the minus 1.5 volt voltage regulator used in the Computer, Gun Direction: M18, for field artillery applications.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18; Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army Munitions Command

F10525755	Voltage Regulator Subassembly

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PACKAGING DATA SHEET

M14002 Data Sheet, Master Packaging

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication.- The voltage regulator shall be manufactured in accordance with Drawing F10525755 and drawings pertaining thereto.

3.2 General specifications.- The voltage regulator shall meet the following requirements of MIL-F-13926, and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship

3.2.2 MIL-C-46357

a. Etched circuit assembly
b. Marking
c. Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The voltage regulator shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the voltage regulator for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the voltage regulator shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The voltage regulator shall withstand 3 shock impulses in the vertical plane, relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3 Temperatures.

3.3.3.1 Operating.- The voltage regulator shall meet the requirements of 3.4.6 while exposed to temperatures of -40°F and +135°F.

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3.3.3.2 Storage.- The voltage regulator shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

3.3.4 Humidity.

3.3.4.1 Operating.- The voltage regulator shall meet the requirements of 3.4.6 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 135°F.

3.3.4.2 Non-operating.- The voltage regulator shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The voltage regulator shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The voltage regulator shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a fine mist of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Performance.- Unless otherwise specified the voltage regulator shall meet the performance requirements of this specification at ambient temperatures between 60° and 90°F; a barometric pressure of 28 to 32 inches of mercury; and a RH up to 90%. Air shall be circulated across the components of the voltage regulator at the rate of at least 100 feet per minute at the ambient temperatures.

3.4.1 Selection of resistors.- Wiring and connections shall be in accordance with the electrical schematic Drawing F10525758. Resistors listed in Table I shall be selected and assembled on the voltage regulator prior to meeting the subsequent requirements.

TABLE I

Resistors Used to Set Outputs

Function	Resistors
+150V Output Voltage	R71, R72
+15V Output Voltage	R60, R61
-1.5V Output Voltage	R8, R9
+150V Trip Current	R67, R68
+15V Trip Current	R43, R44

3.4.2 Resistors.- After final assembly onto the voltage regulator, resistors which are not functionally part of the circuits referenced in 3.4.6 shall possess characteristics within tolerance limits specified on the appropriate drawings.

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3.4.3 Capacitors.- After final assembly onto the voltage regulator, capacitors which are not functionally part of the circuits referenced in 3.4.6 shall possess characteristics within tolerance limits specified on the appropriate drawings.

3.4.4 Diodes.- After final assembly onto the voltage regulator, diodes which are not functionally part of the circuits referenced in 3.4.6 shall possess characteristics within tolerance limits specified on the appropriate drawings.

3.4.5 Transistors and silicon controlled rectifiers.- After final assembly onto the voltage regulator, transistors and silicon controlled rectifiers which are not functionally part of the circuits referenced in 3.4.6 shall possess characteristics within tolerance limits specified on the appropriate drawings.

3.4.6 Circuit output.- The circuits in this assembly are part of a regulated power supply and when connected to the appropriate external circuitry shall have Output Voltages as given in Table II when the Voltage Inputs and Loads are those given in the same table.

TABLE II

Inputs, Loads, and Outputs

Voltage Inputs: unless otherwise noted the tolerances are ± 0.5V

Name	Low Line (Volts)	Nominal Line (Volts)	High Line (Volts)
+150V Main	+197.0 ± 2.0	+235.0 ± 2.0	+274.0 ± 2.0
+150V Bias	+197.0 ± 2.0	+235.0 ± 2.0	+274.0 ± 2.0
+ 20V	+ 16.0	+ 19.0	+ 22.5
+ 15V Main	+ 19.8	+ 25.0	+ 29.6
+ 6V	+ 5.5 ± 0.05	+ 6.0 ± 0.05	+ 6.5 ± 0.05
- 12V	- 12.0 ± 1.2	- 12.0 ± 1.2	- 12.0 ± 1.2
- 27V	- 21.5	- 26.0	- 30.7
- 32V	- 26.0	- 31.0	- 37.0

Loads: unless otherwise noted the tolerances are ± 5 percent

Name	Nominal Load (Amperes)	Minimum Load (Amperes)	Maximum Load (Amperes)	Trip Load (Amperes)
+150V Load	0.100	0.052	0.152	0.19 ± 0.02
+ 15V Load	1.11	0.98	1.24	1.7 ± 0.1
-1.5V Load	.71	0.22	1.20	

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Output Voltages

Name	At Terminal with Respect to Common	Nominal Temperature D.C. (Volts)	Nominal Temperature Ripple* (Millivolts)	Extreme Temperature D.C. (Volts)	Extreme Temperature Ripple* (Millivolts)
+150V Output	57A	+ 150 ± 2.32	500	+ 150 ± 7.5	500
+ 15V Output	48B	+15.00 ± 0.30	50	+15.00 ± 0.6	50
-1.5V Output	11B	- 1.50 ± 0.03	20	- 1.50 ± 0.07	20
Tripped Output	+150V Output less than plus 70 volts, plus 15 volts Output less than 2 volts, both with resistors across the output equivalent (± 10%) to the load current before trip.				

*Ripple is herein defined as peak-to-peak (p-p) value of any alternating components of the Output Voltage excluding noise of less than 10 microseconds duration.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.-Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables III, IV and V and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table III shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

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4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the unit rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock.- This test shall be conducted in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the unit shall be shocked in the vertical plane only. The impact (shocks) and "G" value shall be in accordance with 3.3.2. A receiving device, representing the connector receptable that receives the unit in the computer, shall be secured to the horizontal surface of the shock tester. Each end of the receiving device shall be clamped such that the female receiving slot will hold the unit in a vertical plane. The unit will then be inserted into the receiving device and shocked. Subsequent to imposing the number of impacts required, a visual examination shall be made for damages or defects to determine compliance with 3.3.2 and then the unit will be inspected as specified in 4.5.1.3 prior to determining compliance with 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours and the test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 97 $\pm$ 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.8 of this specification, and shall meet the specified requirements of 3.3.4.1.

4.5.1.4 Salt spray.-The unit shall be subjected to the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.4 to 4.5.8.6 inclusive for compliance with 3.3.6.

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4.3.3 Failure of sample.- Should any one item of a standard sampling fail to meet the specified test requirements, acceptance of the represented production lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

Item		Description
1.	Voltage measuring device	0-200vdc, input impedance 10 megohms or greater, accuracy ± .5% or better
2.	Resistance measuring device	0-200 ohms, accuracy ± .5% of scale reading
3.	Current measuring device	0-10 amperes, accuracy ± .1% full scale reading
4.	Oscilloscope	Rise time of scope plus measuring probe, 50 nano seconds or less, sweep rate 50 nano seconds per centimeter, input impedance 8 megohms and 25 pico farads or greater.

In addition to the measuring devices specified above, the voltage regulator subassembly shall utilize the power supplies, signal sources, components and potentiometers outlined in Tables II, VI, VII, and VIII, and Figures 1 through 6.

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TABLE III
(100 percent inspection)

	Characteristic	Requirements	Test Procedures
101.	Selection of resistors	3.4.1	4.5.3 thru 4.5.3.6
102.	Resistors	3.4.2	4.5.4
103.	Capacitors	3.4.3	4.5.5
104.	Diodes	3.4.4	4.5.6
105.	Transistors and silicon controlled rectifiers	3.4.5	4.5.7
106.	Circuit output	3.4.6	4.5.8 thru 4.5.8.6

4.3 Standard sampling.

4.3.1 General.- One voltage regulator shall be selected at random as a standard sample from each 100 produced. The sample shall meet the requirements and tests in Table IV.

TABLE IV

	Characteristic	Requirements	Test Procedures
301.	Fabrication	3.1	Applicable drawings-visual
302.	General specifications	3.2	MIL-F-13926 and MIL-C-46357 - visual

4.3.2 Environmental.- Three units shall be selected at random as a standard sample from each 100 produced or from each month's production, whichever occurs first. The standard samples shall have met the requirements and tests in Table III and shall then meet the requirements and tests in Table V.

TABLE V

	Characteristic	Requirements	Test Procedures
303.	Vibration	3.3.1	4.5.1.1
304.	Shock	3.3.2	4.5.1.2
305.	Humidity - Operating	3.3.4.1, 3,4.6	4.5.1.3.2,
306.	Humidity - Non-operating	3.3.4.2	4.5.1.3.1
307.	Salt spray	3.3.6	4.5.1.4
308.	Operating temperature	3.3.3.1, 3.4.6	4.5.1.5, 4.5.8
309.	Storage temperature	3.3.3.2	4.5.1.5

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4.5.1.5 Temperature (operating and storage).- After being subjected to all environmental tests specified above, the unit shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the unit shall be tested in accordance with 4.5.8 to 4.5.8.6 to determine compliance with the requirements of 3.4.6 as specified in 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.4 to 4.5.8.6 inclusive to determine compliance with 3.3.3.2.

4.5.1.6 Fungus.- One of the first 5 units produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Performance.- To determine compliance with 3.4, the test methods and procedures in 4.5.3 through 4.5.3.6 shall be performed as specified.

4.5.3. Selection of resistors.- To determine compliance with 3.4.1, the resistors listed in Table I shall be selected as indicated herein so that the voltage regulator shall meet the tests of 4.5.3.1 to 4.5.3.6 inclusive. Wiring shall be visually inspected to insure that all components are physically present and conform to Drawing Nos. 10525755 and 10525758. If any of the resistors listed in Table I are missing, they shall be selected to meet the tests of 4.5.3.1 to 4.5.3.6. If all components are present and conform to applicable drawings, the tests of 4.5.8.1 to 4.5.8.6 inclusive shall be performed.

4.5.3.1 External circuitry.- Connect the external circuitry to the voltage regulator as shown in Figures 2, 3, 4, 5, and 6. The external power supply characteristics shall be those of Table VI and Figure 1. Additional conditions for testing shall be as follows:

(a) Potentiometers.- Connect the potentiometers as indicated in Table VII.

(b) Loads.- Adjust all loads listed in Table II such that for nominal output voltages the load current conditions of all regulators shall not exceed those listed in the Maximum Current column or be less than those listed in the Minimum Current column.

(c) Power supply turn-on sequence.- Energize the power supplies. Unless otherwise noted, the following sequency shall always be followed when energizing the power supplies: 1) Energize the minus 12V, minus 27V, and minus 32V power supplies; 2) Energize the plus 20, and plus 6V, power supplies; 3) Energize the plus 150v Main, plus 150v Bias and plus 15v Main Power supplies.

4.5.3.2 Output voltage resistors.- Adjust the potentiometers 1, and 2, and 3, all listed in Table VII, such that the output voltages are as close to those listed in the Output Voltage Nominal Temperature column of Table II as the potentiometer resolution will allow. The output voltages shall be read to an accuracy of ± 0.1 percent. The Voltage Inputs and Loads shall be those listed in the Nominal Line and Nominal Load columns of Table II.

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4.5.3.3 Plus 150V trip current resistors.- At the conditions defined in the Nominal Line and Maximum Load columns of Table II, adjust potentiometer 4 of Table VII such that when the plus 150V Load is increased the voltage regulator trips when the load is as close to 0.19 amperes as the resolution of the potentiometer allows. The Tripped Output is defined under Output Voltages in Table II. The load current shall be measured to an accuracy of ± 4 milliamperes. At no time shall the plus 150V load current exceed 0.23 amperes. After the voltage regulator is tripped, it may be reset by de-energizing all power supplies and, with proper loads, re-energizing them in proper sequency.

4.5.3.4 Plus 15V trip current resistors.- At conditions defined in the Nominal Line and Maximum Load columns of Table II, adjust potentiometer 5 of Table VII such that when the plus 15V load is increased, the voltage regulator trips when the plus 15V Load current is as close to 1.7 amperes as the resolution of the potentiometer setting allows. The current shall be read to an accuracy of ± 25 milliamperes. At no time shall the current exceed 1.9 amperes.

4.5.3.5 Installation of resistors.- Complete the fabrication of the voltage regulator by installing the resistors listed in Table I, using resistors listed in Table VIII. The resistor values shall be those determined by the potentiometer settings for the respective resistors.

4.5.3.6 Resistor selection test.- With the potentiometers of Table VII disconnected and at Nominal Line and Nominal Load column conditions of Table II, the voltage from terminal 15B to 6B shall be minus 1.500 ± 0.015 volts dc.

4.5.4 Resistors.- After final assembly, the resistors shall be tested as specified on the appropriate drawings and shall meet the requirements specified in 3.4.2.

4.5.5 Capacitors.- After final assembly, the capacitors shall be tested as specified on the appropriate drawings and shall meet the requirements specified in 3.4.3.

4.5.6 Diodes.- After final assembly, the diodes shall be tested as specified on the appropriate drawings and shall meet the requirements specified in 3.4.4.

4.5.7 Transistors and silicon controlled rectifiers.- After final assembly, the transistors and silicon controlled rectifiers shall be tested as specified on the appropriate drawing and shall meet the requirements specified in 3.4.5.

4.5.8 Circuit outputs.- To determine compliance with 3.4.6 perform the following tests specified in 4.5.8.1 through 4.5.8.6.

4.5.8.1 External circuitry.- Connect the external circuitry to the voltage regulator as shown in Figures 2, 3, 4, 5, and 6. The power supply characteristics shall be those of Table VI and Figure 1. Additional conditions for testing shall be as follows:

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TABLE VII

POTENTIOMETER CONNECTIONS

Potentiometer	Quantity Set	Type*	+ Terminal	- Terminal	Wiper Term.	Initial Wiper Setting From + Terminal (ohms)
1	+150V Output Voltage	III	45A	33A	39B	55 ohms
2	+15V Output Voltage	II	13B	14B	5B	35 ohms
3	-1.50V Output Voltage	I	2A	3A	5A	15 ohms
4	+150V Trip Current	III	14A	10A	8A	105 ohms
5	+15V Trip Current	III	19A	18A	24A	105 ohms

Type*	Total Resistance ohms (± 1 percent)		Required Range of Wiper ohms	Current Rating amperes
I	30	1	1.7 to 28.4	.04
II	70	1	5 to 65	.04
III	110	1	5 to 105	.04

All types shall be capable of wiper settings in increments of 5 ohms ± 5 percent over the required wiper range except the Type I potentiometer whose increments shall be 1.7 ohms ± 5 percent. The potentiometer shall be used in such a way that the wiper is never open circuited. (The total resistance shall be within the 1 percent tolerance specified at all times when measurements are being made but may have a ± 6 ohms tolerance at other times).

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(a) Loads.- Adjust all loads in Table II such that for nominal output voltages the load current conditions shall not exceed those listed in the Maximum Load column or be less than those listed in the Minimum Load column.

(b) Power supply turn-on sequence.- Energize the power supplies. Unless otherwise noted, the following sequence shall always be followed when energizing the power supplies: 1) Energize the minus 12V, minus 27V and minus 32V power supplies; 2) Energize the plus 20, and plus 6V power supplies; 3) Energize the plus 150V Main and plus 15V Main power supplies.

4.5.8.2 Output voltage.- The output voltages of the voltage regulator shall be the Output Voltage Nominal Temperature column of Table II when the conditions of the Voltage Inputs and Loads of Table II are combined in the following column pairs:

(a) Low Line - Maximum Load
(b) Low Line - Minimum Load
(c) High Line - Minimum Load
(d) High Line - Maximum Load

4.5.8.3 Plus 150V overload trip current.- At conditions defined in the Low Line and Maximum Load columns of Table II, increase the plus 150V load. The voltage regulator shall trip at the plus 150V Trip Load. The plus 150V Load shall never exceed 0.19 amperes. The voltage regulator Tripped Output is defined under Output Voltage in Table II. The voltage from terminal 26A to 58A shall be 0 to plus 2 volts. With the voltage regulator in the tripped condition, perform the following:

(a) Place a shorted ammeter from terminal 57A to common and remove the short. The current shall not exceed 0.085 amperes. Remove the ammeter. After the voltage regulator is tripped, it may be reset by de-energizing all power supplies and, with proper loads, re-energizing them in proper sequence.

(b) Reset the voltage regulator. At Nominal Line and Maximum Load conditions of Table II, the voltage from 57A to common shall be 150.00 ± 2.32 volts.

4.5.8.4 Plus 15V overload trip current.- At conditions defined in the Low Line and Maximum Load columns of Table II, increase the plus 15V load current. The voltage regulator shall trip at the plus 15V Trip Load of Table II. The current shall not exceed 1.9 amperes. The voltage from 19B to 2B shall be 0 to plus 2.0 volts.

4.5.8.5 Marginal test.- Perform the following at the conditions in the Nominal Line and Nominal Load columns of Table II;

(a) Disconnect terminal 24B from common. The plus 15V output shall increase in magnitude 5.0 ± 1.0 percent. Reconnect terminal 24B.

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(b) Connect terminals 18B to common. The plus 15V output shall decrease in magnitude 5.0 ± 1.0 percent. Disconnect terminal 18B from common.

4.5.8.6 Remote sensing.- At conditions defined in the Nominal Line and Nominal Load columns of Table II, disconnect terminals 6B, 15B, 20B, and 17B. No output voltage shall change more than 0.5 volts from the output voltage measured when these terminals are connected. Reconnect the terminals.

TABLE VI

POWER SUPPLIES

Three phase full wave bridge types*

Identifying Name	Variable Range of Output (Volts)	Nominal Setting (Volts)	Current Rating (Amperes)
+150V Main	+197 to +274	+235	0.25
+150V Bias	+197 to +274	+235	0.10
+ 15V Main	+ 19 to + 30	+ 25	2.0
+ 20V	+ 16 to + 28	+21.8	2.0
- 32V	- 23 to - 40	-32.0	0.15
- 27V	- 23 to - 34	-29.0	0.1

Regulated types

Identifying Name	Variable Range of Output (Volts)	Max. RMS Ripple (Millivolts)	Current Rating (Amperes)
+ 6V P.S.	+ 5 to + 6	2	0.1
-12V P.S.	-11 to -13	2	0.1

* See Figure 1 for characteristics of this type power supplies.

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TABLE VIII

RESISTORS

Resistor Numbers	Value Ohms
10526597	1.7
10526701	3.4
10526702	5
10526703	6
10526704	6.7
10526705	8.4
10526706	10
10526707	11.7
10526708	13.4
10526709	15
10526710	16.7
10526711	18.4
10526712	20
10526713	21.7
10526714	23.4
10526715	25
10526716	26.7
10526717	28.4
10526718	30
10526719	35
10526720	40
10526721	45
10526722	50
10526723	55
10526724	60
10526725	65
10526726	70
10526727	75
10526728	80
10526729	85
10526730	90
10526731	95
10526732	100
10526733	105

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with Master Packaging Data Sheet M14002.

6. NOTES

6.1 Intended use.- The voltage regulator contains the control elements of a plus 150 volt series type voltage regulator, a minus 1.50 shunt type regulator, and a plus 15 volt series type voltage regulator. Provisions are made for fail-safe remote sensing at the load and for increasing or decreasing the plus 15 volt output voltage by five percent for marginal testing procedures. Temperature compensated overcurrent protection is provided for the plus 150 volt and plus 15 volt regulators. When overcurrent conditions are detected, a signal is sent to an external master trip circuit that shuts down selected regulators.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet (see 5.1).
(e) A block of serial numbers to identify each voltage regulator.

Custodian:

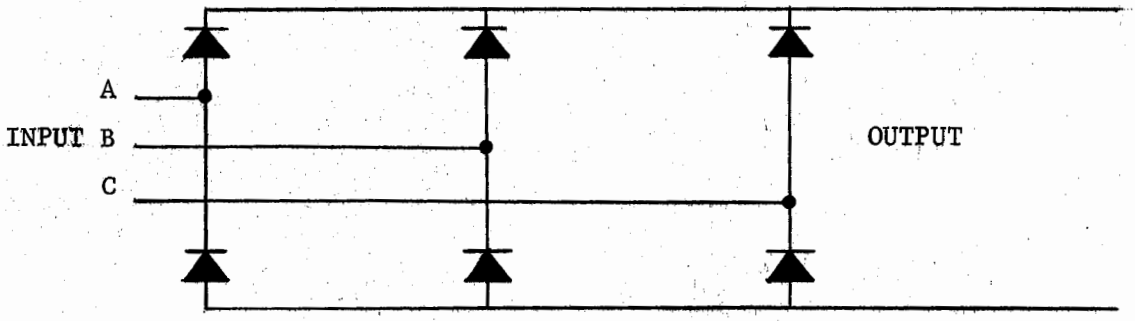
Army - MU

Preparing activity:

Army - MU

Project No. 1220-A194Y

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INPUT: Frequency: 400 ± 20 cycles per second

Line Balance: The voltage variation between any two lines shall not exceed 8 percent of the average of the three line to line voltages.

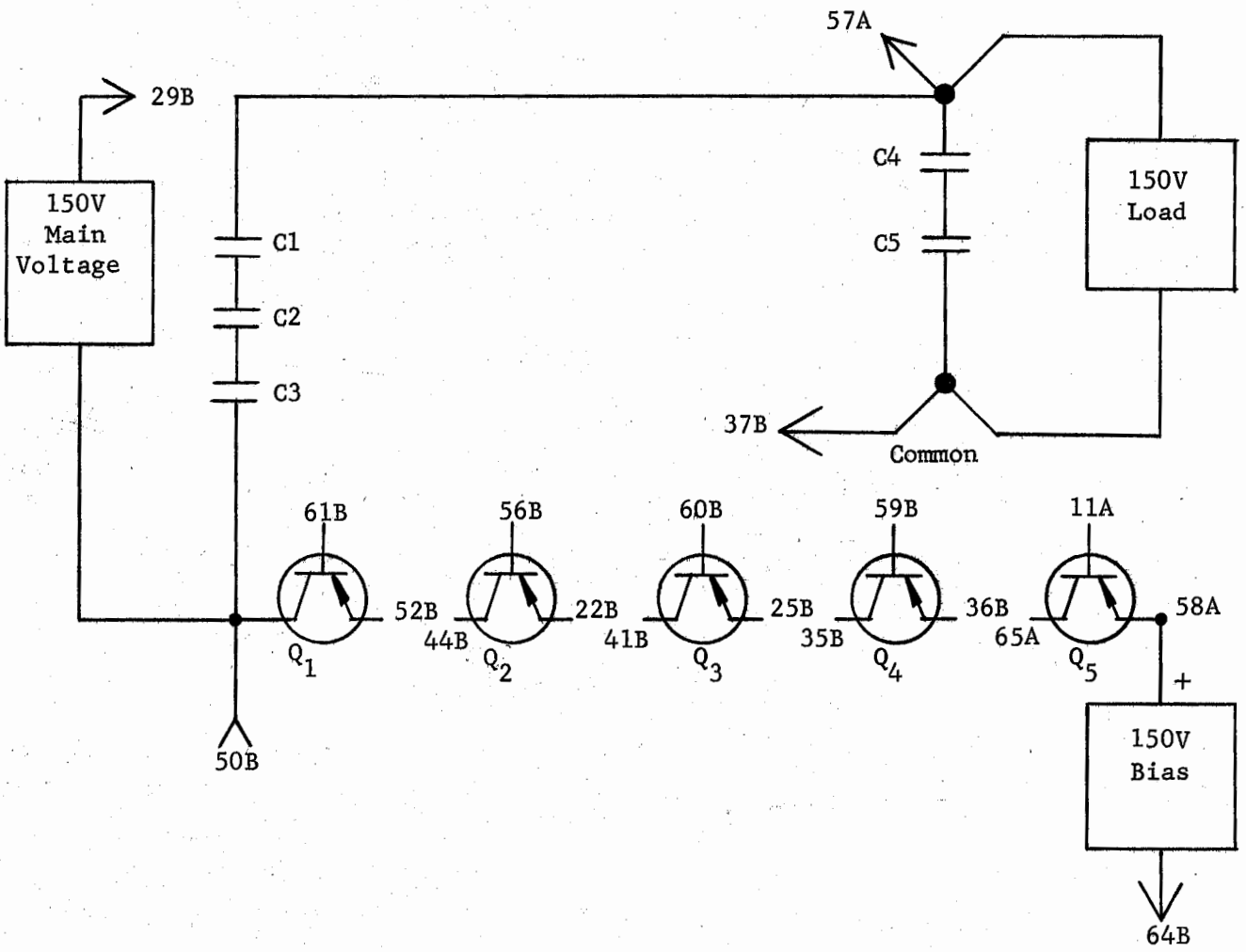
OUTPUT The peak to peak (p-p) ripple* of the output shall not exceed 20 percent of the average direct current output voltage

* Ripple is herein defined as peak-to-peak (p-p) value of any alternating component of the Output Voltage excluding noise of less than 10 micro-second duration.

FIGURE 1

Three Phase Full Wave Bridge

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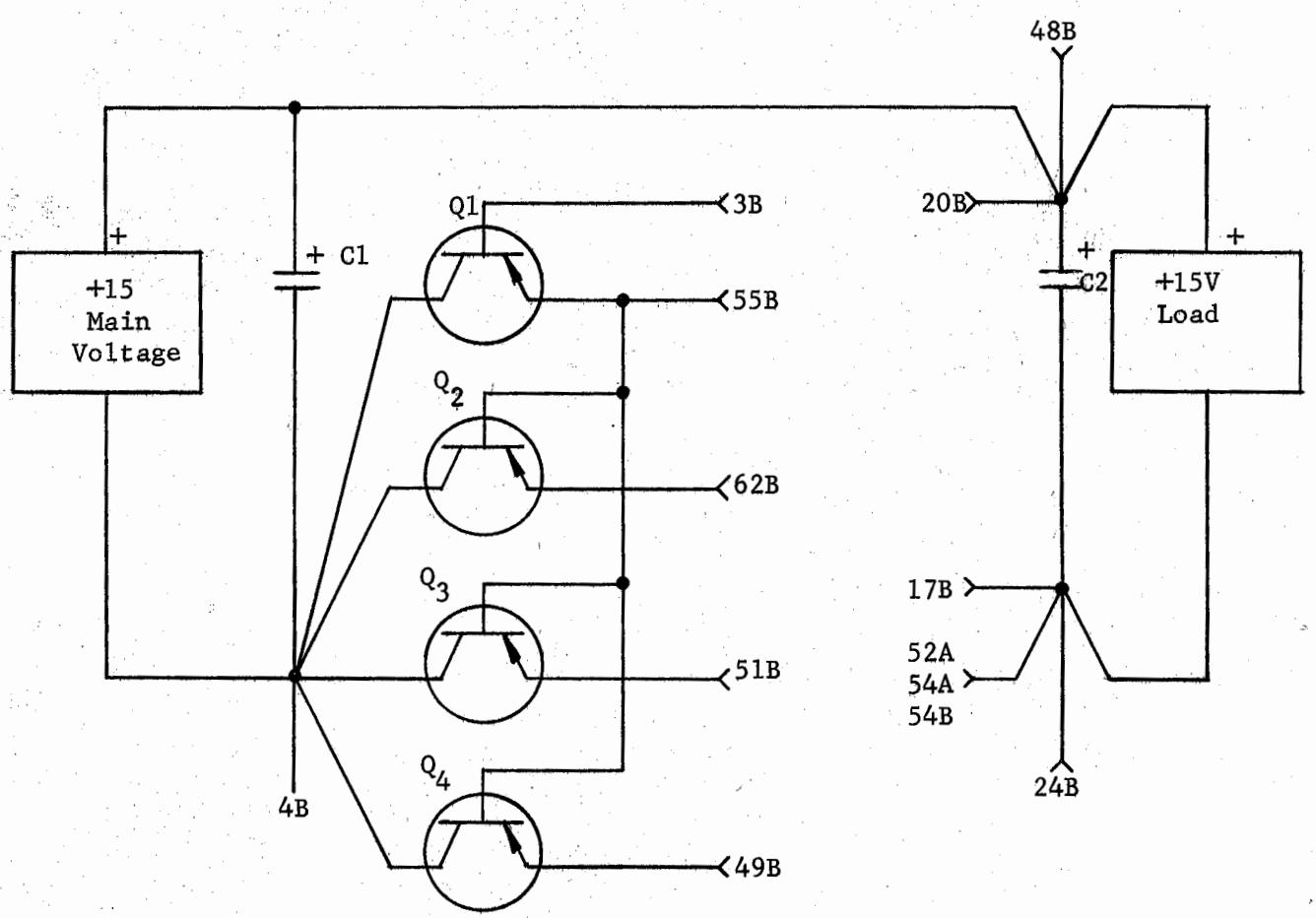
All Transistors: 10525749 Transistor-Power

All Capacitors: 10525815 Capacitor-Tantalytic

Terminal numbers refer to those on the Regulator No. 2 Assembly

FIGURE 2. External Circuitry: +150V Regulator

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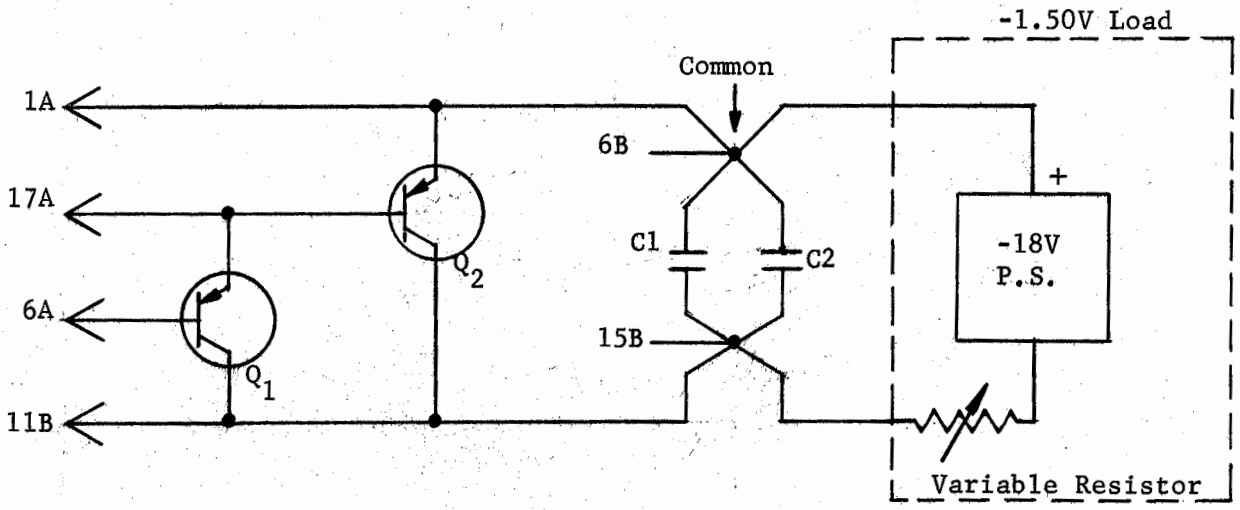


Q1:	10525764	Transistor-Medium Power
Q2, Q3, Q4:	10525749	Transistor-Power
C1:	10525817	Capacitor-Tantalytic
C2:	10525816	Capacitor-Tantalytic

Terminal numbers refer to those on the Regulator No. 2 Assembly

FIGURE 3. External Circuitry: +15V Regulator

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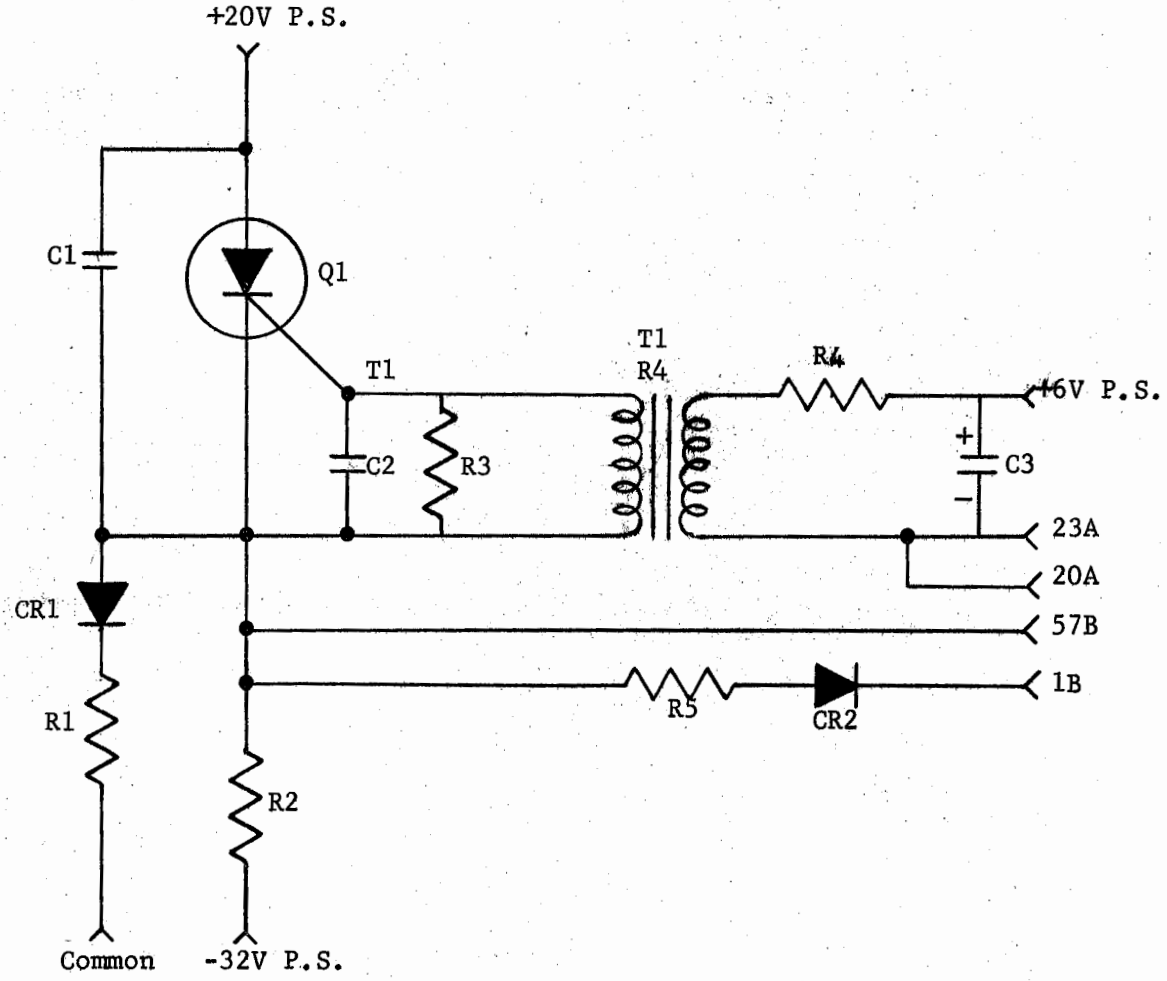
Q1:	10525764	Transistor - Medium Power
Q2:	10525749	Transistor - Power
C1, C2:	10525819	Capacitor - Tantalytic

The -1.50V Load is equivalent to an active current source consisting of a -18V power supply with a maximum RMS Ripple of 25 MV in series with a variable resistor.

Terminal numbers refer to those of the Regulator No. 2 Assembly

FIGURE 4 External Circuit: -1.5V Regulator

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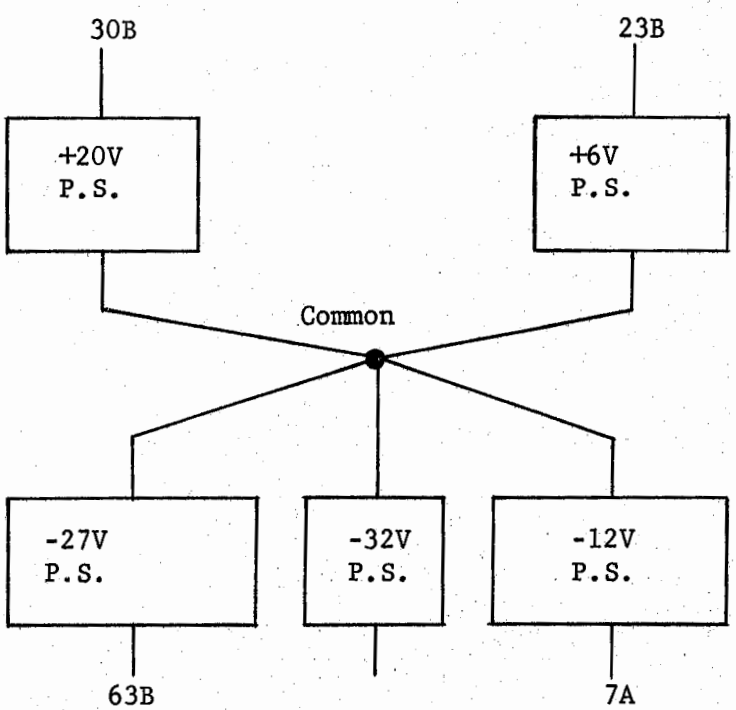
Q1: 10525779 Rectifier, Silicon Controlled
C1: 10525829 Capacitor-Mylar
C2: 10525810 Capacitor-Fusion, Sealed
C3: 10525824 Capacitor-Fixed, Solid Electrolyte
R1: 10525778-5 Resistor-Power, Wirewound
R2: 10526649 Resistor
R3: MS35043-87 Resistor
R4: 10526660 Resistor-Wirewound
R5: 10525778-18 Resistor-Power, Wirewound
CR1: 10525737 Diode-Silicon, Medium Power
CR2: C9144277 Diode
T1: 10525744 Transformer-Pulse

Terminals refer to those on the Regulator No. 2 Assembly.

FIGURE 5

EXTERNAL CIRCUITRY: Master Trip Circuit

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Terminal numbers refer to those on the Regulator No. 2 Assembly

FIGURE 6 External Circuit: Power Supplies

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R0

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in curement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information the use of this specification which will insure that suitable products can be procured with a minimum amount of delay at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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